

Risk and AI (RAI) Practice Exam

100 Questions — Modules 3, 4, and 5

Balanced Options with Full Explanations

Original Practice Set

October 10, 2026

1. Module 3: Fairness, Explainability, and Human Risk

Question 1

A medical AI diagnoses a rare disease affecting 0.1% of the population. Untreated, the disease is almost always fatal. Which metric best assesses the algorithm's performance?

- A. Sensitivity, which measures the proportion of actual positives correctly identified.
- B. Precision, which measures the proportion of predicted positives that are correct.
- C. Accuracy, which measures the proportion of all predictions that are correct.
- D. Specificity, which measures the proportion of actual negatives correctly identified.

Correct Answer: A

Explanation: A is correct. Sensitivity (true positive rate) captures the proportion of actual positive cases correctly identified. Missing a true case is fatal. B is incorrect. Precision ignores false negatives, which are the critical error in this scenario. C is incorrect. Accuracy is misleading with rare events because a never-positive classifier scores 99.9%. D is incorrect. Specificity captures true negatives, which matter less when the disease is curable with few side effects.

Learning Objective: Describe various measures of group fairness. **Reference:** Module 3

Question 2

How does demographic parity contribute to group fairness?

- A. It ensures that model accuracy is equal across all subgroups.
- B. It ensures that predicted positive rates are equal across subgroups.
- C. It ensures that model sensitivity is equal across all subgroups.
- D. It ensures that model precision is equal across all subgroups.

Correct Answer: B

Explanation: A is incorrect. Demographic parity says nothing about accuracy. B is correct. Demographic parity requires equal predicted positive rates. C is incorrect. Equal sensitivity is a different metric (equal opportunity). D is incorrect. Precision equality is not required by demographic parity.

Learning Objective: Describe various measures of group fairness. **Reference:** Module 3

Question 3

A fraud model is trained on 10,000 transactions, only 20 of which are fraudulent. Which form of algorithmic bias is most likely to occur?

- A. Measurement bias, arising from inconsistent feature measurement.
- B. Representation bias, arising from poor subgroup representation.
- C. Data composition bias, arising from severe class imbalance.
- D. Historical bias, arising from outdated training patterns.

Correct Answer: C

Explanation: A is incorrect. Measurement bias concerns how features are measured. B is incorrect. Representation bias concerns subgroup under-representation. C is correct. Severe class imbalance (20 vs. 9,980) is composition bias. D is incorrect. Historical bias concerns outdated data, not imbalance.

Learning Objective: Describe sources of algorithmic bias. **Reference:** Module 3

Question 4

A hiring model uses education level. To prevent education from influencing predictions, the best approach is to:

- A. Remove the education variable entirely from the trained model.
- B. Replace education with its average value during the scoring step.
- C. Add interaction terms between education and other model features.
- D. Replace education with binary dummy variables for each degree.

Correct Answer: B

Explanation: A is incorrect. Removing the variable leaves its influence embedded in correlated features. B is correct. Setting education to its average equalizes treatment at prediction time. C and D are incorrect. They change representation but do not remove influence.

Learning Objective: Describe sources of algorithmic bias. **Reference:** Module 3, Section 2.3.1

Question 5

A bank's chatbot sometimes gives incorrect information. The most appropriate action is:

- A. Collect additional customer inputs before generating each answer.
- B. Develop an ethical framework and periodically review outputs.
- C. Generate periodic risk reports summarizing chatbot performance.
- D. Implement design mechanisms requiring human verification of outputs.

Correct Answer: D

Explanation: A is incorrect. More inputs do not prevent hallucination. B is incorrect. A framework alone does not fix incorrect responses. C is incorrect. Reporting describes but does not mitigate the problem. D is correct. Human verification prevents incorrect outputs from reaching customers.

Learning Objective: Discuss risks posed by AI to human autonomy. **Reference:** Module 3

Question 6

To increase a fraud model's explainability, a credit card company should:

- A. Provide clear reasoning for the decisions produced by the model.
- B. Use a deep neural network to improve predictive performance.
- C. Provide access to the source code, data, and design rationale.
- D. Design the model with simple features and linear relationships.

Correct Answer: A

Explanation: A is correct. Explainability = understandable reasoning for decisions. B is incorrect. Deep networks reduce explainability. C is incorrect. Code/data access increases transparency, not explainability. D is incorrect. Simplicity increases interpretability, not explainability.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 7

Customers complain they do not understand why loan applications are rejected. The source of reputational risk and its mitigation is:

- A. A privacy breach, mitigated by securing customer data access.
- B. Algorithmic bias, mitigated by using more diverse training data.
- C. Lack of explainability, mitigated by providing rejection reasons.
- D. Lack of transparency, mitigated by disclosing all training data.

Correct Answer: C

Explanation: A and B are incorrect. Neither is described. C is correct. The complaint is about unexplained rejections. D is incorrect. Disclosing training data is inappropriate and insufficient.

Learning Objective: Describe sources of AI-related reputational risk. **Reference:** Module 3

Question 8

A technology company wants to minimize data-related risk. The best step is:

- A. Retain all customer data regardless of sensitivity level.
- B. Give customers the ability to transfer or delete their data.
- C. Store customer data with multiple third-party vendors.
- D. Maintain an unencrypted original copy in a secure database.

Correct Answer: B

Explanation: A, C, D are incorrect. These increase data risk. B is correct. Data subject rights (transfer, deletion) reduce abuse.

Learning Objective: Describe ethical principles of privacy. **Reference:** Module 4

Question 9

Which best distinguishes explainability from interpretability?

- A. Explainability and interpretability are essentially identical.
- B. Interpretability concerns model structure; explainability concerns individual predictions.
- C. Explainability applies only to linear models; interpretability is universal.
- D. Interpretability requires deep networks; explainability requires shallow models.

Correct Answer: B

Explanation: A is incorrect. They are related but distinct. B is correct. Interpretability = model design; explainability = prediction explanation. C and D are incorrect. They reverse the actual relationship.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 10

A model trained on majority-group data performs worse for a minority group. This is primarily:

- A. Measurement bias, caused by inconsistent feature collection.
- B. Representation bias, caused by subgroup under-representation.
- C. Historical bias, caused by outdated training patterns.
- D. Label bias, caused by incorrect target labeling.

Correct Answer: B

Explanation: A is incorrect. Measurement bias concerns feature measurement. B is correct. Under-representation causes poor subgroup performance. C and D are incorrect. Not indicated by the scenario.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 11

Which action best supports procedural fairness in an AI lending system?

- A. Publishing the model's source code for public inspection.
- B. Providing a mechanism for applicants to contest decisions.
- C. Increasing the model's complexity to improve accuracy.
- D. Removing all applicant data after a decision is rendered.

Correct Answer: B

Explanation: A is incorrect. Code publication supports transparency. B is correct. Contest mechanisms support procedural fairness. C and D are incorrect. Unrelated to procedural fairness.

Learning Objective: Discuss the principles of justice. **Reference:** Module 4

Question 12

A model's decision threshold is lowered to reduce false negatives. This will most likely:

- A. Increase precision by filtering out more false positives.
- B. Increase recall by capturing more actual positive cases.
- C. Decrease sensitivity by reducing the true positive rate.
- D. Decrease recall by missing more actual positive cases.

Correct Answer: B

Explanation: A is incorrect. Lowering the threshold usually reduces precision. B is correct. More positives captured = higher recall. C and D are incorrect. Recall and sensitivity both increase.

Learning Objective: Evaluate classification model performance. **Reference:** Module 2, Chapter 8

Question 13

A bank's AI rejects a loan without giving specific reasons. Which principle is most directly violated?

- A. Autonomy, which concerns respecting individual decision-making.
- B. Non-maleficence, which concerns avoiding harm to others.
- C. Explainability, which concerns providing reasons for decisions.
- D. Beneficence, which concerns acting in the interest of others.

Correct Answer: C

Explanation: A, B, D are incorrect. Secondary principles in this scenario. C is correct. Failure to explain a decision is an explainability failure.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 14

"Model interpretability" is most important when:

- A. The model is used internally for exploratory data analysis.
- B. Decisions affect individuals and require formal justification.
- C. The model is trained primarily on synthetic or generated data.
- D. The model has no exposure to regulatory or legal review.

Correct Answer: B

Explanation: A, C, D are incorrect. These contexts lower the need for interpretability. B is correct. High-stakes individual decisions require justification.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 15

Which is the best example of a human-in-the-loop safeguard?

- A. Fully automated loan approval with periodic post-hoc audits.
- B. Human review of AI-flagged AML alerts before regulatory filing.
- C. Fully automated trading with post-trade reconciliation checks.
- D. Auto-generation of regulatory filings with quarterly review.

Correct Answer: B

Explanation: A, C, D are incorrect. These are largely automated processes. B is correct. Human review before consequential action = HITL.

Learning Objective: Discuss risks posed by AI to human autonomy. **Reference:** Module 3

Question 16

A face recognition system performs poorly on darker skin tones. This is most likely caused by:

- A. Sampling bias, resulting from under-representation in training data.
- B. Excessive model capacity, resulting in overfitting to fine details.
- C. Over-regularization, resulting in loss of model flexibility.
- D. Insufficient feature engineering, resulting in poor feature maps.

Correct Answer: A

Explanation: A is correct. Subgroup under-representation causes subgroup failure. B, C, D are incorrect. Not the primary cause.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 17

Which is the most appropriate response when an AI hiring model shows disparate impact?

- A. Ignore the finding because overall model accuracy is high.
- B. Audit features, retrain with balanced data, and document findings.
- C. Deploy the model and monitor subgroup performance annually.
- D. Replace the model with a purely random candidate selection.

Correct Answer: B

Explanation: A is incorrect. Overall accuracy does not excuse disparate impact. B is correct. Audit, retrain, and document. C is incorrect. Annual monitoring is too slow for an active issue. D is incorrect. Random selection destroys utility.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 18

A loan model uses zip code as a feature, producing discriminatory outcomes. This illustrates:

- A. Proxy discrimination, where a neutral feature encodes a protected one.
- B. Concept drift, where the input-output relationship changes over time.
- C. Overfitting, where the model memorizes noise in the training data.
- D. Data drift, where the distribution of input features changes over time.

Correct Answer: A

Explanation: A is correct. Zip code acts as a proxy for race and other protected attributes. B and D are incorrect. Temporal phenomena not indicated. C is incorrect. Overfitting is unrelated.

Learning Objective: Describe sources of algorithmic bias. **Reference:** Module 3

Question 19

Which fairness metric requires equal false positive rates across groups?

- A. Demographic parity, which equalizes positive prediction rates.
- B. Equalized odds, which equalizes both TPR and FPR across groups.
- C. Predictive parity, which equalizes positive predictive value.
- D. Calibration, which aligns predicted probabilities with outcomes.

Correct Answer: B

Explanation: A is incorrect. Demographic parity concerns positive prediction rates. B is correct. Equalized odds requires equal TPR and FPR. C and D are incorrect. Different fairness definitions.

Learning Objective: Describe various measures of group fairness. **Reference:** Module 3

Question 20

A regulator asks for the reasoning behind a credit denial. The bank cannot provide one. This most directly exposes the bank to:

- A. Market risk, arising from adverse price movements in positions.
- B. Liquidity risk, arising from an inability to meet cash obligations.
- C. Conduct and reputational risk, arising from unexplained decisions.
- D. Settlement risk, arising from failed counterparty transactions.

Correct Answer: C

Explanation: A, B, D are incorrect. Unrelated exposures. C is correct. Failure to explain decisions triggers conduct and reputational exposure.

Learning Objective: Describe sources of AI-related reputational risk. **Reference:** Module 3

Question 21

Which is the best approach to address historical bias in training data?

- A. Remove the protected attribute from the training dataset.
- B. Re-weight samples, audit outcomes, and document remediation.
- C. Increase the model's complexity to capture more patterns.
- D. Restrict training data to only the most recent observations.

Correct Answer: B

Explanation: A is incorrect. Removing the attribute does not remove the bias. B is correct. Re-weighting, auditing, and documenting is standard. C and D are incorrect. Not effective remediations.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 22

A model shows high accuracy overall but poor recall for defaults. This is most concerning because:

- A. Accuracy is misleading in datasets with significant class imbalance.
- B. Precision is always more important than recall in credit models.

- C. Recall is only meaningful for continuous regression targets.
- D. Accuracy is a fairness metric that ignores class labels.

Correct Answer: A

Explanation: A is correct. High accuracy can mask poor minority-class recall. B is incorrect. Recall is often more important. C is incorrect. Recall is a classification metric. D is incorrect. Accuracy is not a fairness metric.

Learning Objective: Evaluate classification model performance. **Reference:** Module 2, Chapter 8

Question 23

An AI system used in hiring produces outcomes that disadvantage a protected group. Under most regulatory frameworks, the firm must:

- A. Take no action if overall accuracy is high for all candidates.
- B. Document, remediate, and demonstrate non-discriminatory outcomes.
- C. Publicly disclose the source code and training data for the model.
- D. Replace the model entirely with purely human hiring decisions.

Correct Answer: B

Explanation: A is incorrect. Accuracy does not excuse discrimination. B is correct. Documentation, remediation, and demonstration. C and D are incorrect. Not required.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 24

Which best describes “fairness through unawareness”?

- A. Removing protected attributes from the model’s feature set.
- B. Training the model only on protected attributes and outcomes.
- C. Using demographic data as the sole basis for decisions.
- D. Ignoring all fairness metrics during model development.

Correct Answer: A

Explanation: A is correct. Fairness through unawareness = omitting protected attributes. B, C, D are incorrect. They are the opposite or irrelevant.

Learning Objective: Describe various measures of group fairness. **Reference:** Module 3

Question 25

The primary purpose of an AI ethics review board is to:

- A. Approve model deployment without requiring further review.
- B. Provide independent oversight of ethical implications.
- C. Approve marketing materials for AI-related products.
- D. Set interest rates for AI-driven credit products.

Correct Answer: B

Explanation: A is incorrect. The board reviews, does not rubber-stamp. B is correct. Independent ethical oversight. C and D are incorrect. Outside the board’s remit.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 26

A model's performance differs across age groups. This is:

- A. Concept drift, caused by a change in the input-output relationship.
- B. Disparate impact, caused by differential outcomes across groups.
- C. Overfitting, caused by excessive model complexity and memorization.
- D. Underfitting, caused by insufficient model capacity to fit the data.

Correct Answer: B

Explanation: A is incorrect. Concept drift is temporal. B is correct. Different outcomes across groups = disparate impact. C and D are incorrect. Training issues, not group-specific.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 3

Question 27

Which best mitigates bias in an AI credit scoring system?

- A. Use only alternative data sources to replace traditional features.
- B. Combine fairness testing, diverse data, and human oversight.
- C. Use deeper neural networks to capture more complex patterns.
- D. Remove all features that correlate with protected attributes.

Correct Answer: B

Explanation: A is incorrect. Alternative data introduces its own bias. B is correct. Multi-layered mitigation. C is incorrect. Deep networks can amplify bias. D is incorrect. Removing features destroys utility.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 28

Which is a human autonomy risk of AI in finance?

- A. Reduced transaction costs from automated processing.
- B. Decisions made without meaningful human review.
- C. Improved regulatory reporting through automation.
- D. Faster settlement of securities transactions.

Correct Answer: B

Explanation: A, C, D are incorrect. These are benefits, not autonomy risks. B is correct. Loss of meaningful review threatens autonomy.

Learning Objective: Discuss risks posed by AI to human autonomy. **Reference:** Module 3

Question 29

A chatbot gives medical advice beyond its intended scope. This most directly creates:

- A. Market risk, from adverse movements in traded positions.
- B. Conduct risk, from inappropriate customer interactions.
- C. Liquidity risk, from an inability to meet cash demands.
- D. Interest rate risk, from adverse rate movements.

Correct Answer: B

Explanation: A, C, D are incorrect. Unrelated risks. B is correct. Out-of-scope advice is a conduct issue.

Learning Objective: Discuss risks posed by AI to human autonomy. **Reference:** Module 3

Question 30

Which is the best way to increase trust in an AI system?

- A. Publish the model weights for full public inspection.
- B. Provide transparent explanations and clear audit trails.
- C. Increase model complexity to improve predictive accuracy.
- D. Reduce testing frequency to speed up deployment cycles.

Correct Answer: B

Explanation: A is incorrect. Weights alone do not build trust. B is correct. Transparency and audit trails build trust. C and D are incorrect. Reduce trust.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 31

Which is a valid criticism of using accuracy alone to evaluate a fairness-sensitive model?

- A. Accuracy is not a legitimate performance metric in classification.
- B. Accuracy can hide poor performance for under-represented groups.
- C. Accuracy applies only to regression models, not classifiers.
- D. Accuracy ignores the class labels used in training the model.

Correct Answer: B

Explanation: A, C, D are incorrect. Factually wrong. B is correct. Accuracy can mask subgroup failures.

Learning Objective: Evaluate classification model performance. **Reference:** Module 2, Chapter 8

Question 32

Which AI risk is most associated with deepfakes in financial markets?

- A. Interest rate risk, from adverse movements in benchmark rates.
- B. Manipulation and misinformation risk, from fabricated content.
- C. Credit spread risk, from deterioration in issuer credit quality.
- D. Sovereign risk, from adverse government policy changes.

Correct Answer: B

Explanation: A, C, D are incorrect. Unrelated. B is correct. Deepfakes enable market manipulation and misinformation.

Learning Objective: Discuss risks posed by AI to human autonomy. **Reference:** Module 3

Question 33

A model's explanations are post-hoc approximations. This means:

- A. The explanations are mathematically exact and fully faithful.
- B. The explanations approximate the model's reasoning, not its true logic.
- C. The model produces no outputs that can be explained at all.
- D. The model is underfitting and therefore unreliable.

Correct Answer: B

Explanation: A is incorrect. Post-hoc explanations are approximate. B is correct. They approximate rather than reveal true logic. C and D are incorrect. Unrelated to post-hoc explanation.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 34

Which framework best supports organizational AI ethics?

- A. Ad-hoc committees convened only when issues arise.
- B. Documented principles, review processes, and accountability.
- C. Outsourcing all ethical decisions to external vendors.
- D. Avoiding documentation to preserve proprietary methods.

Correct Answer: B

Explanation: A, C, D are incorrect. Not robust frameworks. B is correct. Documented principles + review + accountability.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 35

Which is the best action when a deployed AI produces unexpected outcomes for a subgroup?

- A. Ignore the issue if overall metrics remain strong.
- B. Escalate, investigate, and document remediation steps.
- C. Delete the affected logs to avoid regulatory exposure.
- D. Increase the model size to improve subgroup performance.

Correct Answer: B

Explanation: A is incorrect. Overall metrics can hide subgroup issues. B is correct. Escalate, investigate, document. C is incorrect. Deleting logs is a governance failure. D is incorrect. Model size is irrelevant.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

2. Module 4: Ethics, Privacy, and Principles

Question 36

Which statement correctly describes Utilitarianism?

- A. It can lead to violations of individual rights for the greater good.
- B. It focuses on virtuous character traits and living a good life.
- C. It relies heavily on laws, regulations, and duty-based rules.
- D. It values the intrinsic morality of actions over their outcomes.

Correct Answer: A

Explanation: A is correct. Aggregate welfare can justify violating individual rights. B is incorrect. That describes virtue ethics. C is incorrect. That describes deontology. D is incorrect. That also describes deontology.

Learning Objective: Compare ethical frameworks. **Reference:** Module 4

Question 37

A deontological framework judges morality based on:

- A. The consequences of the action for total social welfare.
- B. Adherence to ethical duties and rules regardless of outcome.
- C. A single, quantifiable metric for overall utility.
- D. Virtuous character traits and the pursuit of a good life.

Correct Answer: B

Explanation: A is incorrect. Consequences = consequentialism. B is correct. Deontology = duty-based. C is incorrect. Utility metric = consequentialism. D is incorrect. Virtue = virtue ethics.

Learning Objective: Compare ethical frameworks. **Reference:** Module 4

Question 38

Regarding the principle of justice in AI ethics:

- A. Different concepts of distributive justice can conflict with one another.
- B. Impartial application of the law is optional for procedural justice.
- C. Personal autonomy is generally secondary to achieving social justice.
- D. Meritocracy is not considered a principle of distributive justice.

Correct Answer: A

Explanation: A is correct. Egalitarianism and utilitarianism can conflict. B is incorrect. Impartial application is required. C is incorrect. Social justice requires autonomy. D is incorrect. Meritocracy is a distributive principle.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 39

An AI recommends treatment for an unconscious patient that conflicts with documented preferences. Which principle is violated?

- A. Explainability, which concerns clarity of the AI's reasoning.
- B. Beneficence, which concerns acting in the patient's interest.

- C. Autonomy, which concerns respecting the patient's choices.
- D. Non-maleficence, which concerns avoiding harm to the patient.

Correct Answer: C

Explanation: A, B, D are incorrect. Secondary in this scenario. C is correct. Ignoring documented preferences violates autonomy.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 40

A model trained on data from the US and Canada performs poorly in Brazil, Nigeria, and Thailand. The most likely cause is:

- A. Bias in problem specification, from a mismatch in target variables.
- B. Bias in model deployment, from a failure in the rollout process.
- C. Historical bias, from outdated patterns in the training data.
- D. Sampling bias, from a non-representative training population.

Correct Answer: D

Explanation: A is incorrect. Not about target variables. B is incorrect. Not about deployment. C is incorrect. Not about time. D is correct. Non-representative training sample.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 41

A model trained on 2005–2015 data underperforms in 2026 because societal patterns changed. This is:

- A. Sampling bias, from a mismatch between training and deployment populations.
- B. Historical bias, from using outdated data reflecting past conditions.
- C. Measurement bias, from inconsistent data collection methods over time.
- D. Data composition bias, from an imbalance in the training dataset.

Correct Answer: B

Explanation: A is incorrect. Not about geographic representation. B is correct. Outdated data = historical bias. C and D are incorrect. Not indicated.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 42

Which ethical principle requires avoiding harm?

- A. Beneficence, which concerns acting for others' benefit.
- B. Non-maleficence, which concerns avoiding harm to others.
- C. Justice, which concerns fair distribution of benefits.
- D. Autonomy, which concerns respecting individual decisions.

Correct Answer: B

Explanation: A is incorrect. Beneficence = act for benefit. B is correct. Non-maleficence = do no harm. C and D are incorrect. Different principles.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 43

A company sells customer data without consent. Which principle is most directly violated?

- A. Explainability, which concerns clarity of decisions made by AI.
- B. Privacy, which concerns control over personal information.
- C. Fairness, which concerns equal treatment across groups.
- D. Transparency, which concerns disclosure of model design.

Correct Answer: B

Explanation: A, C, D are incorrect. Secondary or unrelated. B is correct. Selling data without consent is a privacy violation.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 44

Which best describes virtue ethics?

- A. An approach that focuses on consequences for overall welfare.
- B. An approach that focuses on duties and rules above outcomes.
- C. An approach that focuses on character traits and living well.
- D. An approach that focuses on maximizing total utility.

Correct Answer: C

Explanation: A is incorrect. Consequences = consequentialism. B is incorrect. Duties = deontology. C is correct. Virtue ethics = character. D is incorrect. Utility = consequentialism.

Learning Objective: Compare ethical frameworks. **Reference:** Module 4

Question 45

Which is the primary concern of beneficence in AI ethics?

- A. Avoiding harm to individuals and groups affected by AI systems.
- B. Acting in the interest of others and promoting their wellbeing.
- C. Preserving individual autonomy in decisions involving AI systems.
- D. Distributing benefits and burdens fairly across all groups.

Correct Answer: B

Explanation: A is incorrect. Avoiding harm = non-maleficence. B is correct. Beneficence = act for benefit. C is incorrect. Autonomy = self-determination. D is incorrect. Justice = fair distribution.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 46

Which best describes the principle of autonomy?

- A. Respecting individuals' right to make their own decisions.
- B. Maximizing aggregate utility across all affected parties.
- C. Following established rules regardless of consequences.
- D. Acting in a virtuous manner consistent with good character.

Correct Answer: A

Explanation: A is correct. Autonomy = self-determination. B is incorrect. Utility = consequentialism. C is incorrect. Rules = deontology. D is incorrect. Virtue = virtue ethics.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 47

A firm uses AI to make lending decisions without human review. Which principle is most directly at risk?

- A. Non-maleficence, which concerns avoiding harm to borrowers.
- B. Autonomy, which concerns respecting borrowers' decision-making.
- C. Beneficence, which concerns acting in borrowers' interest.
- D. Justice, which concerns fair treatment of all borrowers.

Correct Answer: B

Explanation: A, C, D are incorrect. Secondary. B is correct. No human review threatens applicant autonomy.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 48

Which best describes informational privacy?

- A. Control over how personal data is collected and used.
- B. The right to physical security and bodily integrity.
- C. Ownership of intellectual property created by individuals.
- D. Protection of freedom of speech and expression.

Correct Answer: A

Explanation: A is correct. Informational privacy = control over personal data. B, C, D are incorrect. Different concepts.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 49

Which is a legitimate basis for processing personal data under most privacy frameworks?

- A. Any business interest that the firm deems important.
- B. Explicit consent or a legal obligation to process.
- C. Speculation about future uses of the data.
- D. A general agreement with a third-party vendor.

Correct Answer: B

Explanation: A, C, D are incorrect. Not sufficient legal bases. B is correct. Consent or legal obligation.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 50

Which best describes data minimization?

- A. Collecting as much data as possible to enable future use.
- B. Collecting only data necessary for the stated purpose.
- C. Storing data indefinitely to preserve analytical value.
- D. Sharing data with all vendors that support the business.

Correct Answer: B

Explanation: A, C, D are incorrect. Contradict minimization. B is correct. Minimization = collect only what is needed.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 51

A firm retains customer data longer than necessary. This violates:

- A. Data minimization, which limits collection and retention.
- B. Autonomy, which concerns respecting individual decisions.
- C. Beneficence, which concerns acting in others' interest.
- D. Transparency, which concerns disclosure of data practices.

Correct Answer: A

Explanation: A is correct. Retention beyond necessity violates minimization. B, C, D are incorrect. Secondary or unrelated.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 52

Which best describes deontology?

- A. An ethical framework based on the consequences of actions.
- B. An ethical framework based on duties and moral rules.
- C. An ethical framework based on character and virtue.
- D. An ethical framework based on maximizing total utility.

Correct Answer: B

Explanation: A is incorrect. Consequence-based = consequentialism. B is correct. Duty-based = deontology. C is incorrect. Character = virtue ethics. D is incorrect. Utility = consequentialism.

Learning Objective: Compare ethical frameworks. **Reference:** Module 4

Question 53

Which best describes consequentialism?

- A. Actions are judged by their outcomes and overall effects.
- B. Actions are judged by adherence to duties and rules.
- C. Actions are judged by the character of the actor.
- D. Actions are judged by their conformity to laws.

Correct Answer: A

Explanation: A is correct. Consequentialism = outcomes. B is incorrect. Duties = deontology. C is incorrect. Character = virtue ethics. D is incorrect. Laws = legal positivism.

Learning Objective: Compare ethical frameworks. **Reference:** Module 4

Question 54

A firm uses AI to determine credit limits based on social media activity. The key ethical concern is:

- A. Privacy and consent, from using non-financial personal data.
- B. Interest rate levels, from adverse rate movements.
- C. Model inference speed, from computational constraints.
- D. Data storage cost, from increased storage requirements.

Correct Answer: A

Explanation: A is correct. Social media data raises consent/privacy concerns. B, C, D are incorrect. Unrelated to the ethics issue.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 55

Which is a risk of using alternative data in credit models?

- A. Only improved inclusion of previously unscorable applicants.
- B. Potential use of sensitive or non-consented personal data.
- C. Reduced model accuracy relative to traditional data.
- D. Lower regulatory scrutiny of the credit decision process.

Correct Answer: B

Explanation: A is incorrect. Inclusion is a potential benefit, not the only effect. B is correct. Alternative data may be sensitive or non-consented. C is incorrect. Accuracy may improve. D is incorrect. More scrutiny, not less.

Learning Objective: Describe data governance frameworks. **Reference:** Module 5

Question 56

Which best describes the principle of transparency?

- A. Publishing model weights and full training code publicly.
- B. Providing accessible information about how decisions are made.
- C. Sharing all training data with affected individuals.
- D. Publishing source code without documentation.

Correct Answer: B

Explanation: A, C, D are incorrect. Specific actions that may not constitute transparency. B is correct. Transparency = accessible information on decisions.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 57

Which is the best approach when ethical principles conflict?

- A. Ignore the conflict and proceed with the fastest option.
- B. Apply a documented prioritization and governance framework.
- C. Follow the option that maximizes short-term profit.
- D. Choose arbitrarily to avoid delaying deployment.

Correct Answer: B

Explanation: A, C, D are incorrect. Not ethical responses. B is correct. Documented prioritization and governance.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 58

Which best describes the role of consent in AI?

- A. Consent is optional in all cases involving personal data.
- B. Consent is required when no other legal basis exists.
- C. Consent is never required for commercial AI applications.
- D. Consent is only required for government AI applications.

Correct Answer: B

Explanation: A, C, D are incorrect. Consent is generally required absent legal basis. B is correct. Consent required if no other legal basis.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 59

Which best describes algorithmic fairness?

- A. Ensuring identical outcomes for every individual in a population.
- B. Ensuring fair treatment across groups based on chosen criteria.
- C. Ensuring maximum accuracy regardless of subgroup performance.
- D. Ensuring no automated model is used in high-stakes decisions.

Correct Answer: B

Explanation: A is incorrect. Identical outcomes are not always the goal. B is correct. Fair treatment based on chosen fairness criteria. C is incorrect. Accuracy alone is not fairness. D is incorrect. Not a definition of fairness.

Learning Objective: Describe various measures of group fairness. **Reference:** Module 3

Question 60

Which best describes non-maleficence?

- A. Do no harm to individuals and groups affected by AI.
- B. Do good by acting in the interest of others.
- C. Act fairly by distributing benefits equitably.
- D. Respect autonomy by honoring individual choices.

Correct Answer: A

Explanation: A is correct. Non-maleficence = do no harm. B is incorrect. Beneficence = do good. C is incorrect. Justice = act fairly. D is incorrect. Autonomy = respect choice.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 61

Which best describes beneficence?

- A. Acting in a way that benefits others and promotes wellbeing.
- B. Avoiding harm to individuals and groups affected by AI.
- C. Respecting individuals' rights to make their own decisions.
- D. Distributing benefits and burdens fairly across all groups.

Correct Answer: A

Explanation: A is correct. Beneficence = act for benefit. B is incorrect. Avoid harm = non-maleficence. C is incorrect. Respect rights = autonomy. D is incorrect. Fair distribution = justice.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 62

A firm uses AI to make employment decisions but keeps the algorithm secret. This most directly affects:

- A. Transparency, which concerns accessible information about decisions.
- B. Autonomy, which concerns respecting individual decision-making.
- C. Beneficence, which concerns acting in others' interest.
- D. Non-maleficence, which concerns avoiding harm to others.

Correct Answer: A

Explanation: A is correct. Secrecy undermines transparency. B, C, D are incorrect. Secondary.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 63

Which best describes distributive justice?

- A. Fair allocation of benefits and burdens across a society.
- B. Fair application of rules and procedures to all.
- C. Fair development of individual character and virtue.
- D. Fair maximization of aggregate utility and welfare.

Correct Answer: A

Explanation: A is correct. Distributive justice = fair allocation. B is incorrect. Fair rules = procedural. C is incorrect. Character = virtue ethics. D is incorrect. Utility = consequentialism.

Learning Objective: Discuss the principles of justice. **Reference:** Module 4

Question 64

Which best describes retributive justice?

- A. Fair punishment for wrongdoing proportional to the offense.
- B. Fair allocation of resources across all societal groups.
- C. Fair procedural application of rules and laws.
- D. Fair development of virtuous character in individuals.

Correct Answer: A

Explanation: A is correct. Retributive = just punishment. B is incorrect. Allocation = distributive. C is incorrect. Procedure = procedural. D is incorrect. Character = virtue ethics.

Learning Objective: Discuss the principles of justice. **Reference:** Module 4

Question 65

Which best describes procedural justice?

- A. Fair processes for making and enforcing decisions.
- B. Fair outcomes in the distribution of benefits.
- C. Fair development of individual character traits.
- D. Fair maximization of overall social welfare.

Correct Answer: A

Explanation: A is correct. Procedural justice = fair processes. B is incorrect. Outcomes = distributive. C, D are incorrect. Unrelated.

Learning Objective: Discuss the principles of justice. **Reference:** Module 4

Question 66

Which is the best response to a privacy breach caused by AI?

- A. Ignore the breach if the number of affected users is small.
- B. Notify affected parties, remediate, and document the incident.

- C. Delete the affected logs to prevent further disclosure.
- D. Publicly blame users for sharing their personal information.

Correct Answer: B

Explanation: A, C, D are incorrect. Irresponsible and potentially illegal. B is correct. Notification + remediation + documentation.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 67

Which best describes data subject rights?

- A. The rights of individuals over their own personal data.
- B. The rights of firms over customer data they collect.
- C. The rights of regulators to access corporate data.
- D. The rights of vendors to use data shared by clients.

Correct Answer: A

Explanation: A is correct. Data subject = the individual. B, C, D are incorrect. Not the standard definition.

Learning Objective: Describe principles of privacy. **Reference:** Module 4

Question 68

Which is a potential harm of AI in hiring?

- A. Reduced bias in hiring decisions through automation.
- B. Amplified historical discrimination against certain groups.
- C. Increased salaries for all candidates who apply.
- D. Lower recruiting costs for the hiring organization.

Correct Answer: B

Explanation: A is incorrect. Bias is not automatically reduced. B is correct. Historical discrimination can be amplified. C, D are incorrect. These are not harms.

Learning Objective: Discuss sources of algorithmic bias. **Reference:** Module 4

Question 69

Which best describes the principle of accountability in AI?

- A. No single party is responsible for AI outcomes.
- B. Clear responsibility for AI outcomes and their consequences.
- C. Only the developers are responsible for AI outcomes.
- D. Only the end users are responsible for AI outcomes.

Correct Answer: B

Explanation: A, C, D are incorrect. Incomplete or wrong. B is correct. Accountability = clear responsibility.

Learning Objective: Discuss the principles of AI ethics. **Reference:** Module 4

Question 70

Which best describes the ethical concern with autonomous AI decision-making?

- A. Loss of human oversight and clear accountability for decisions.
- B. Increased transparency and explainability of decisions.

- C. Reduced operating cost from automation of decisions.
- D. Improved fairness from consistent algorithmic treatment.

Correct Answer: A

Explanation: A is correct. Loss of oversight and accountability is the concern. B, C, D are incorrect. These are benefits, not concerns.

Learning Objective: Discuss risks posed by AI to human autonomy. **Reference:** Module 3

3. Module 5: Model Risk Management and Governance

Question 71

A local bank periodically validates its VaR model. Which is a best practice?

- A. Validation should always occur on a fixed quarterly schedule.
- B. Validation should assess whether original assumptions remain valid.
- C. Validation is discretionary and not required by regulators.
- D. Validation should be conducted by the bank's CRO personally.

Correct Answer: B

Explanation: A is incorrect. Frequency depends on model risk ranking. B is correct. Validation checks assumptions, data, and performance. C is incorrect. Validation is regulatorily prescribed. D is incorrect. No such requirement.

Learning Objective: Discuss model validation and its importance. **Reference:** Module 5

Question 72

A CRO evaluates the model landscape. The most crucial consideration is:

- A. Ensuring that each model operates fully independently of others.
- B. Ensuring that all models use the same set of input data.
- C. Ensuring that models can be consolidated to reduce complexity.
- D. Ensuring that interdependencies between models are identified.

Correct Answer: D

Explanation: A is incorrect. Independence is not required. B is incorrect. Same inputs are not required. C is incorrect. Consolidation is not the focus. D is correct. Interdependencies are the focus of landscape review.

Learning Objective: Describe factors for model inventory. **Reference:** Module 5

Question 73

Which correctly describes appropriate roles in model governance?

- A. The board of directors should maintain the model inventory.
- B. The model risk function should set the model risk appetite.
- C. Model ownership should never be shared between individuals.
- D. Internal and external auditors are necessary independent checks.

Correct Answer: D

Explanation: A is incorrect. Model risk function maintains the inventory. B is incorrect. The board sets model risk appetite. C is incorrect. Ownership may be shared. D is correct. Auditors provide independent checks.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 74

Which team ensures a new ML model integrates with IT systems without disrupting operations?

- A. The Core Implementation Team, which handles model algorithms.
- B. The System Implementation Team, which handles system integration.
- C. The Model Design Team, which handles model architecture.
- D. The Data Management Team, which handles data pipelines.

Correct Answer: B

Explanation: A, C, D are incorrect. Not their primary responsibility. B is correct. System Implementation Team integrates the model.

Learning Objective: Describe steps in model implementation. **Reference:** Module 5

Question 75

What is an advantage of increasing model monitoring frequency in rapidly changing environments?

- A. It reduces the need for model retraining over time.
- B. It detects performance degradation or drift for timely intervention.
- C. It minimizes the need for IT involvement in model operations.
- D. It improves the model's interpretability to business users.

Correct Answer: B

Explanation: A is incorrect. Retraining is still required. B is correct. Monitoring detects drift early. C is incorrect. Monitoring may increase IT involvement. D is incorrect. Monitoring does not change interpretability.

Learning Objective: Describe AI/ML model review framework. **Reference:** Module 5

Question 76

Relative to traditional statistical models, data quality requirements for AI/ML models are:

- A. Identical, because both require the same data standards.
- B. Less rigorous, because AI/ML models are more robust to noise.
- C. Focused only on outlier detection and removal procedures.
- D. More stringent, because AI/ML models use alternative data sources.

Correct Answer: D

Explanation: A, B, C are incorrect. AI/ML models have higher requirements. D is correct. AI/ML data quality requirements are more stringent.

Learning Objective: Describe data governance frameworks. **Reference:** Module 5

Question 77

A new AI mortgage prepayment model outperforms the existing statistical model in testing. The appropriate next step is:

- A. Deploy the model directly to production for immediate use.
- B. Notify the IT department to roll out the model to users.
- C. Retire the existing model and replace it with the new one.
- D. Request initial validation from the model validation team.

Correct Answer: D

Explanation: A, C are incorrect. Deployment requires validation first. B is incorrect. IT notification belongs to the design phase. D is correct. Initial validation is required before production.

Learning Objective: Discuss model validation and its importance. **Reference:** Module 5, Section 3.2.1

Question 78

A new AI credit model uses school transcripts, part-time jobs, social media, and landlord-tenant disputes. A validation finding should be:

- A. No material concerns, because augmented data improves inclusion.
- B. Material concerns about sensitive or non-consented alternative data.
- C. Approval, because backtesting shows improvement over the old model.
- D. Recommendation to include additional alternative data sources.

Correct Answer: B

Explanation: A is incorrect. Sensitive data raises concerns. B is correct. Sensitive alternative data should trigger material concerns. C is incorrect. Backtesting does not override data ethics concerns. D is incorrect. More alternative data increases risk.

Learning Objective: Describe data governance frameworks. **Reference:** Module 5

Question 79

Which is a typical element of a data governance framework for AI?

- A. Data minimization as the sole governing principle.
- B. Provenance, quality, access controls, and lifecycle management.
- C. Vendor contract terms as the primary governance mechanism.
- D. Cloud storage cost controls and usage optimization.

Correct Answer: B

Explanation: A, C, D are incorrect. Incomplete elements. B is correct. Provenance, quality, access, and lifecycle are core.

Learning Objective: Describe data governance frameworks. **Reference:** Module 5

Question 80

Model owners in complex organizations:

- A. Must always be a single individual to ensure accountability.
- B. May be shared among individuals or assigned to a committee.
- C. Cannot be assigned to committees under any circumstances.
- D. Are always external parties independent of the organization.

Correct Answer: B

Explanation: A, C, D are incorrect. Ownership may be shared. B is correct. Shared ownership is allowed in complex organizations.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 81

Which is not a typical step in AI/ML model implementation?

- A. Model design, including architecture and feature selection.
- B. System integration with existing IT infrastructure.
- C. Ongoing monitoring after deployment into production.
- D. Permanent exemption from validation once deployed.

Correct Answer: D

Explanation: A, B, C are incorrect. These are typical steps. D is correct. Permanent exemption from validation is not a step.

Learning Objective: Describe steps in model implementation. **Reference:** Module 5

Question 82

Model risk appetite for a bank should be set by:

- A. The model risk function, based on technical analysis.
- B. The board of directors, as part of overall governance.
- C. The CRO alone, without board involvement.
- D. Internal audit, as an independent function.

Correct Answer: B

Explanation: A, C, D are incorrect. Not the correct authority. B is correct. The board sets model risk appetite.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 83

A model inventory should be maintained by:

- A. The board of directors as part of strategic oversight.
- B. The model risk function as part of its mandate.
- C. Internal audit as part of its independent review.
- D. IT operations as part of system administration.

Correct Answer: B

Explanation: A, C, D are incorrect. Not the maintaining function. B is correct. Model risk function maintains the inventory.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 84

The primary objective of periodic model validation is to:

- A. Confirm that the model continues to generate profit for the firm.
- B. Ensure assumptions remain valid and performance is as expected.
- C. Replace the model with a newer version on an annual cycle.
- D. Satisfy internal audit requirements for documentation.

Correct Answer: B

Explanation: A, C, D are incorrect. Not the objective. B is correct. Validation ensures assumptions and performance.

Learning Objective: Discuss model validation and its importance. **Reference:** Module 5

Question 85

Which is a risk unique to generative AI in risk management workflows?

- A. Interest rate exposure from adverse movements in rates.
- B. Hallucination producing fabricated facts and citations.
- C. Liquidity mismatch from timing differences in cash flows.
- D. Settlement risk from failed counterparty transactions.

Correct Answer: B

Explanation: A, C, D are incorrect. Not unique to GenAI. B is correct. Hallucination is a GenAI-specific risk.

Learning Objective: Describe AI/ML model review framework. **Reference:** Module 5

Question 86

Under the EU AI Act, credit scoring of individuals is classified as:

- A. Prohibited, because it involves personal financial data.
- B. High-risk, because it affects access to essential services.
- C. Limited-risk, because it only affects individual consumers.
- D. Minimal-risk, because it is a routine banking operation.

Correct Answer: B

Explanation: A is incorrect. Not prohibited. B is correct. Credit scoring is high-risk. C and D are incorrect. Too low.

Learning Objective: Describe regulatory frameworks for AI. **Reference:** Module 5

Question 87

A bank uses an LLM to summarize counterparty news for credit decisions. The most significant risk is:

- A. Hallucinated facts influencing the resulting credit decision.
- B. High electricity consumption from running the model.
- C. Volatility in token pricing affecting operational cost.
- D. The model's training data cut-off date being outdated.

Correct Answer: A

Explanation: A is correct. Hallucination is the dominant risk. B, C, D are incorrect. Secondary operational concerns.

Learning Objective: Describe AI/ML model review framework. **Reference:** Module 5

Question 88

Third-party AI risk primarily concerns:

- A. Vendor contract cost and pricing renegotiation exposure.
- B. Concentration, data leakage, model opacity, and dependency.
- C. Vendor equity price volatility affecting partnership value.
- D. Vendor exposure to changes in benchmark interest rates.

Correct Answer: B

Explanation: A, C, D are incorrect. Too narrow. B is correct. Multidimensional third-party risk.

Learning Objective: Discuss vendor and third-party risk. **Reference:** Module 5

Question 89

A model's predictions become less accurate over time because input distributions shift. This is:

- A. Concept drift, caused by changes in the input-output relationship.
- B. Data drift, caused by changes in the distribution of inputs.
- C. Overfitting, caused by excessive complexity and memorization.
- D. Underfitting, caused by insufficient model capacity.

Correct Answer: B

Explanation: A is incorrect. Concept drift = change in $P(y|X)$. B is correct. Data drift = change in $P(X)$. C, D are incorrect. Training issues.

Learning Objective: Describe AI/ML model review framework. **Reference:** Module 5

Question 90

Which best describes the relationship between interpretability and explainability?

- A. They are essentially identical concepts used interchangeably.
- B. Interpretability concerns model structure; explainability concerns individual predictions.
- C. Explainability applies only to linear models; interpretability is universal.
- D. Interpretability requires deep networks; explainability requires shallow models.

Correct Answer: B

Explanation: A is incorrect. Related but distinct. B is correct. Interpretability = model structure; explainability = individual predictions. C, D are incorrect. Reverse the relationship.

Learning Objective: Describe explainability, interpretability, and transparency. **Reference:** Module 3

Question 91

Which is the most appropriate governance response when a model's performance degrades significantly?

- A. Continue using it while documenting performance in quarterly reports.
- B. Trigger review, investigate root cause, and remediate or retire.
- C. Increase marketing efforts to compensate for lower performance.
- D. Ignore the degradation if the model was previously validated.

Correct Answer: B

Explanation: A, C, D are incorrect. Irresponsible. B is correct. Trigger review, investigate, remediate or retire.

Learning Objective: Describe steps in model adaptation. **Reference:** Module 5

Question 92

Which best describes the three lines of defense in model risk management?

- A. Business, model risk, internal audit.
- B. Marketing, sales, and human resources.
- C. IT, HR, and finance functions.
- D. Legal, compliance, and marketing.

Correct Answer: A

Explanation: A is correct. Business = first line; model risk = second; audit = third. B, C, D are incorrect. Not the three lines.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 93

Which is the primary responsibility of the first line of defense in model risk?

- A. Independent validation of models before production use.
- B. Owning and managing model risk in day-to-day operations.
- C. Auditing the model governance framework independently.
- D. Setting regulatory policy for model risk management.

Correct Answer: B

Explanation: A is incorrect. Validation = second line. B is correct. First line owns and manages model risk. C is incorrect. Audit = third line. D is incorrect. Policy setting is not first line's role.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 94

Which is the primary responsibility of the second line of defense?

- A. Model development and feature engineering.
- B. Independent model validation and oversight.
- C. Internal audit of the governance framework.
- D. Marketing of AI-based products.

Correct Answer: B

Explanation: A is incorrect. Development = first line. B is correct. Second line = independent validation and oversight. C is incorrect. Audit = third line. D is incorrect. Marketing is unrelated.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 95

Which is the primary responsibility of the third line of defense?

- A. Model development and calibration activities.
- B. Independent audit of the model risk framework.
- C. Model validation and performance testing.
- D. Business use and daily operation of models.

Correct Answer: B

Explanation: A is incorrect. Development = first line. B is correct. Third line = independent audit. C is incorrect. Validation = second line. D is incorrect. Business use = first line.

Learning Objective: Discuss roles and responsibilities in MRM. **Reference:** Module 5

Question 96

A model's documentation is incomplete. This is a violation of:

- A. Model risk management standards requiring documentation.
- B. Accounting standards requiring financial disclosures.
- C. Marketing standards requiring consumer disclosures.
- D. Human resources standards requiring role clarity.

Correct Answer: A

Explanation: A is correct. Documentation is required by MRM standards. B, C, D are incorrect. Unrelated standards.

Learning Objective: Discuss model validation and its importance. **Reference:** Module 5

Question 97

Which is the best approach for validating an AI/ML model?

- A. Only statistical tests on in-sample prediction accuracy.
- B. Conceptual soundness, data quality, outcomes analysis, and monitoring.
- C. Only backtesting against historical performance data.
- D. Only vendor-provided reports and certifications.

Correct Answer: B

Explanation: A, C, D are incorrect. Incomplete approaches. B is correct. Comprehensive validation includes all four elements.

Learning Objective: Discuss model validation and its importance. **Reference:** Module 5